

Rapid optimization of a Buchwald–Hartwig amination using Design of Experiments (*DoE*)



Catalyst screening and DOE optimization for Buchwald-Hartwig amination via Mya4 parallel reaction station

Buchwald–Hartwig couplings are an important tool for the synthesis of aromatic amines. Since its discovery, palladium catalysed cross coupling of amines with aryl halides have been extensively used in medicinal chemistry, materials science, heterocyclics and natural product synthesis. Although, the underlying catalytic process and mechanistic details of C–N couplings have been extensively documented over many years, identifying suitable reaction conditions remains tedious and time consuming.

A major reason for the slow development of Buchwald–Hartwig amination is its sensitive nature towards the catalyst, ligand, additives and even the solvent, thus limiting its reproducibility.

In this study, reaction conditions of C–N coupling were optimized using Radleys Mya4 Reaction Station.

The rapid identification of Buchwald–Hartwig amination conditions was required for the synthesis of building block 3. Initially, a literature review of potential reaction conditions was conducted, and a shortlist of different options was prepared. This shortlist of conditions was subsequently screened on a Radleys Mya 4 Reaction Station using minimum quantities of material, leading to the identification of a set of “hit” conditions. We also screened several greener reaction solvents as alternatives to 1,4-dioxane, leading to the identification of *t*-BuOH as an effective replacement.



Finally, this new process was optimizing using “Design of Experiments” (DoE) software, allowing us to fully explore multiple reaction parameters and variables in as few experiments as possible. In this way, we avoided the fallacy of a “false optimum”, which is often encountered when changing one factor at a time. The desired product was isolated in high yield from our highly optimized process.

Catalyst/base screening

The high throughput screening capabilities of the Radleys Mya 4 Reaction Station allowed for the rapid screening of all sixteen of our shortlisted catalyst- base combinations for this reaction. In this way, a full suite of reaction data was generated, including the UPLC-MS analysis of all 16 reactions, within 2 hours.

The results of this screening process revealed that the use of K_3PO_4 with Xphos Pd G3 gave the highest conversion to product (Table 1). Notably, the use of KO-t-Bu led to bis-addition impurities.

	BrettPhos Pd G4	RuPhos Pd G3	Xphos Pd G3	tBuXPhos Pd G3
KORBu				
K2CO3				
K3PO4				
KOAc				

Table 1: Product conversions

Dioxane solvent swap - safety and sustainability

As a known carcinogen¹, the use of 1,4-dioxane as a reaction solvent raises significant environmental, health and safety (EHS) concerns, highlighting the need for more benign alternatives. A set of greener, sustainable solvents were screened against the best conditions identified from the catalyst-base screening exercise. The solvents used in this experiment were chosen based on well-documented solvent selection guides for greener, more sustainable solvents.^{1,2} Pleasingly, t-BuOH gave comparable results to 1,4-dioxane and was selected for further optimization work (Table 2).

Solvent	% Product Conversion
1,4-Dioxane- Control	88
Toluene	84
2-MeTHF	87
TBME	80
t-BuOH	90
CPME	80
TPGS-750-M	62
DMC	73

Table 2: Alternative solvents screened as potential replacements for 1,4-dioxane

*CPME: Cyclopentyl methyl ether,
TPGS: DL- α -Tocopherol methoxypolyethylene glycol succinate, DMC: Dimethyl carbonate. Reactions conducted in a sealed tube.

Optimum conditions identified when changing one factor at a time: Xphos Pd G3 (5 mol%), K_3PO_4 (3 eq.) t-BuOH (0.4 M conc.), reflux, 2 h – 90% product.

Design of Experiments - DoE

Finally, the reaction was subjected to further optimization using DoE software from Design Expert. This work focused on several key reaction parameters, including catalyst loading, equivalents of base, temperature and concentration. The results of these experiments were used to build a surface response plot (Figure 1), as well as a statistical model of the Buchwald-Hartwig reaction, leading to further improvements in the yield of the reaction (98% yield), as well as avoiding the potential for a “false optimum”, which can be encountered when changing just one factor at a time.

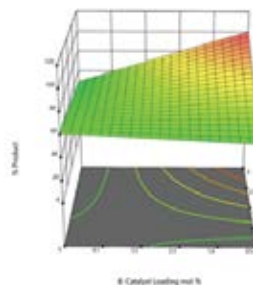


Figure 1: 3D surface response with respect to catalyst loading and equivalency of base.

Conclusion

In summary, the rapid optimization of a Buchwald-Hartwig amination reaction was achieved using the Radleys Mya 4 Reaction Station together with DoE software from Design Expert. Further improvements to the reaction were made by replacing 1,4-dioxane with t-BuOH, which represents a much safer and sustainable reaction solvent.

The use DoE allowed for the generation of a statistical model of the reaction based on multiple factors, as well as the identification of important two- factor interactions. In this way, the model allowed the prediction of optimum conditions for maximizing yield, resulting in 98% conversion to product 3.